

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I M.purushothaman192524517 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and	Good analysis with	Basic analysis with limited	Poor or incorrect analysis of

		validation of results	reasonable interpretation	insights	results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Satellite Internet Communication Using OSI Layers

CSA0716-COMPUTER NETWORKS

UNDER THE GUIDENCE OF :
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INTRODUCTION



- Satellite Internet provides internet access through communication satellites.
- It connects remote and rural areas where wired networks are unavailable. Data is transmitted between user terminals, satellites, and ground stations.
- Common applications include education, disaster recovery, military, and maritime communication.
- Examples: Starlink, HughesNet, Viasat.

PROBLEM STATEMENT

- Limited internet access in remote and rural areas.
- High latency caused by long-distance satellite communication.
- Signal interference due to weather and environmental conditions.
- Need for reliable data transmission and error control.
- Importance of the OSI model in ensuring efficient satellite communication.

OSI Layers interaction with Satellite Internet



Role of Each OSI Layer

Application (Layer 7)	Provides internet services such as web browsing, email, video calls, and streaming.
Presentation (Layer 6)	Encrypts, decrypts, compresses, and formats data before transmission.
Session (Layer 5)	Establishes, manages, and terminates communication sessions between the user and server.
Transport (Layer 4)	Uses TCP/UDP to ensure reliable delivery, flow control, and error recovery.
Network (Layer 3)	Assigns IP addresses and routes packets through satellites and ground stations.
Data Link (Layer 2)	Creates data frames, performs error detection, and controls access to the communication channel.
Physical (Layer 1)	Converts data into radio frequency (RF) signals and transmits them between the satellite dish, satellite, and ground station.

Communication Flow Through OSI Layers

- **Step 1:** User enters a request (website, email, video).
- **Step 2:** Data passes through the **7 OSI layers**:
 - Application → User service
 - Presentation → Encryption & Compression
 - Session → Connection management
 - Transport → TCP/UDP
 - Network → IP Routing
 - Data Link → Framing & Error Detection
 - Physical → RF Signal Transmission
- **Step 3:** Signal travels:
 - User Terminal → Satellite → Ground Station → Internet Server
- **Step 4:** Server response follows the same path back to the user.



AI-BASED SOLUTION



AI-based Traffic Management: Predicts network congestion and selects the best transmission path.

Intelligent Routing: Uses machine learning to optimize packet routing and reduce delays.

Error Detection & Correction: AI identifies transmission errors and minimizes packet loss.

Bandwidth Optimization: Dynamically allocates bandwidth based on user demand and network conditions.

Predictive Maintenance: Detects satellite or ground station faults before failures occur.

Security Enhancement: AI detects cyberattacks, anomalies, and unauthorized access in real time.



APPLICATIONS AND BENEFITS



Applications:

Remote Area Connectivity: Provides internet access in rural and isolated regions.

Disaster Management: Maintains communication during natural disasters and emergencies.

Maritime & Aviation: Enables internet services for ships and aircraft.

Benefits:

Global internet coverage

Reliable communication in remote areas

High-speed data transmission



Conclusion



- Satellite Internet Communication using the OSI model, combined with AI technologies, offers a scalable, secure, and efficient solution for providing reliable global internet connectivity.

THANK YOU